

Source Code for :- BILL SPLITTER & EXPENSE TRACKER

```
class BillSplitter:
    def __init__(self):
        self.users = set()
        self.balances = {}

    def add_user(self, name):
        if name not in self.users:
            self.users.add(name)
            self.balances[name] = {}

            # Initialize balances with other users
            for user in self.users:
                if user != name:
                    self.balances[name][user] = 0
                    self.balances[user][name] = 0
            print(f"User '{name}' added successfully.")
        else:
            print(f"User '{name}' already exists.")

    def add_expense(self, payer, amount, participants):
        if payer not in self.users:
            print(f"Payer '{payer}' not found!")
            return

        for participant in participants:
            if participant not in self.users:
                print(f"Participant '{participant}' not found!")
                return

        split_amount = round(amount / len(participants), 2)

        for participant in participants:
            if participant == payer:
                continue
            self.balances[participant][payer] += split_amount
            self.balances[payer][participant] -= split_amount

        print(f"Expense of ${amount:.2f} added. Each participant owes ${split_amount:.2f} to {payer}.")

    def show_balances(self):
        print("\n----- Balances -----")
        is_empty = True
        for user in self.users:
            for other in self.balances[user]:
                amount = self.balances[user][other]
                if amount > 0:
                    print(f"{user} owes {other}: ${amount:.2f}")
                    is_empty = False
        if is_empty:
            print("No balances to show. All settled up!")

    def simplify(self):
        # Optional: implement debt simplification logic
        pass

def main():
    app = BillSplitter()

    while True:
        print("\n===== Bill Splitter & Expense Tracker =====")
        print("1. Add User")
        print("2. Add Expense")
        print("3. Show Balances")
        print("4. Exit")

        choice = input("Enter your choice (1-4): ")

        if choice == '1':
            name = input("Enter user name: ").strip()
            app.add_user(name)

        elif choice == '2':
            payer = input("Enter payer's name: ").strip()
            try:
                amount = float(input("Enter total amount: "))
            except ValueError:
                print("Invalid amount.")
                continue
            participants = input("Enter participants (comma-separated): ").split(',')
            participants = [p.strip() for p in participants]
            app.add_expense(payer, amount, participants)

        elif choice == '3':
            app.show_balances()

        elif choice == '4':
            print("Exiting. Thank you for using Bill Splitter!")
            break

        else:
            print("Invalid choice. Try again.")

if __name__ == "__main__":
    main()
```